중앙대학교
CHUNG-ANG UNIVERSITY

Chapter 1
Object-Oriented Analysis and Design

이찬근

소프트웨어학부
중앙대학교

Object-Oriented Analysis and Design

OOA & OOD ⟷ Structed-oriented

객체 지향

Analysis tool과 design tool이 호환 X

- **Object-Oriented Analysis** (OOA)
  - Discover the domain concepts (the objects of the problem domain)

Analysis tool과 Design tool이 호환.

(OOD) **Object-Oriented Design**

  - Define software objects and how they collaborate to fulfill the requirements

Book
title

블럭 다이어그램

디자인 다이어그램

Book
- title : String + print() : void

```
public class Book {
    public void print();
    private String title;
}
```

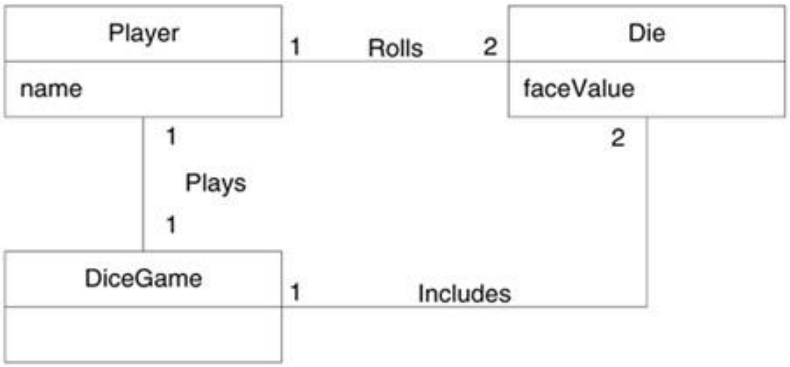
A Short Example of OOAD : "Dice Game"

- **Define Use Cases**

UC1: Play a Dice Game

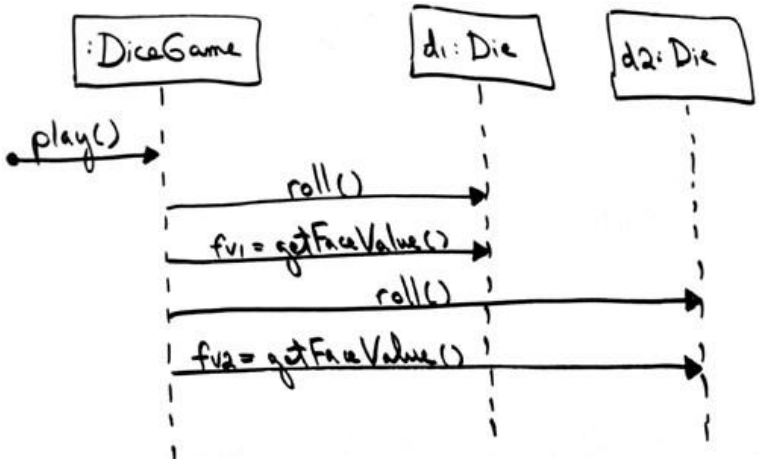
- Step1: Player requests to roll the dice
- Step 2: System presents results
- Step 3: If the dice's face value totals seven, player wins; otherwise, player loses.

- **Define Domain Model**



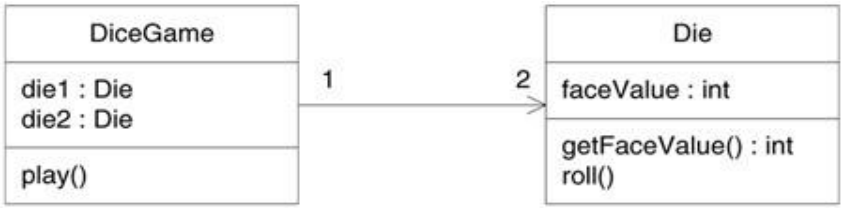
Domain model shows the *noteworthy* domain concepts as objects, their attributes, and associations.

- **Define Interaction Diagrams**

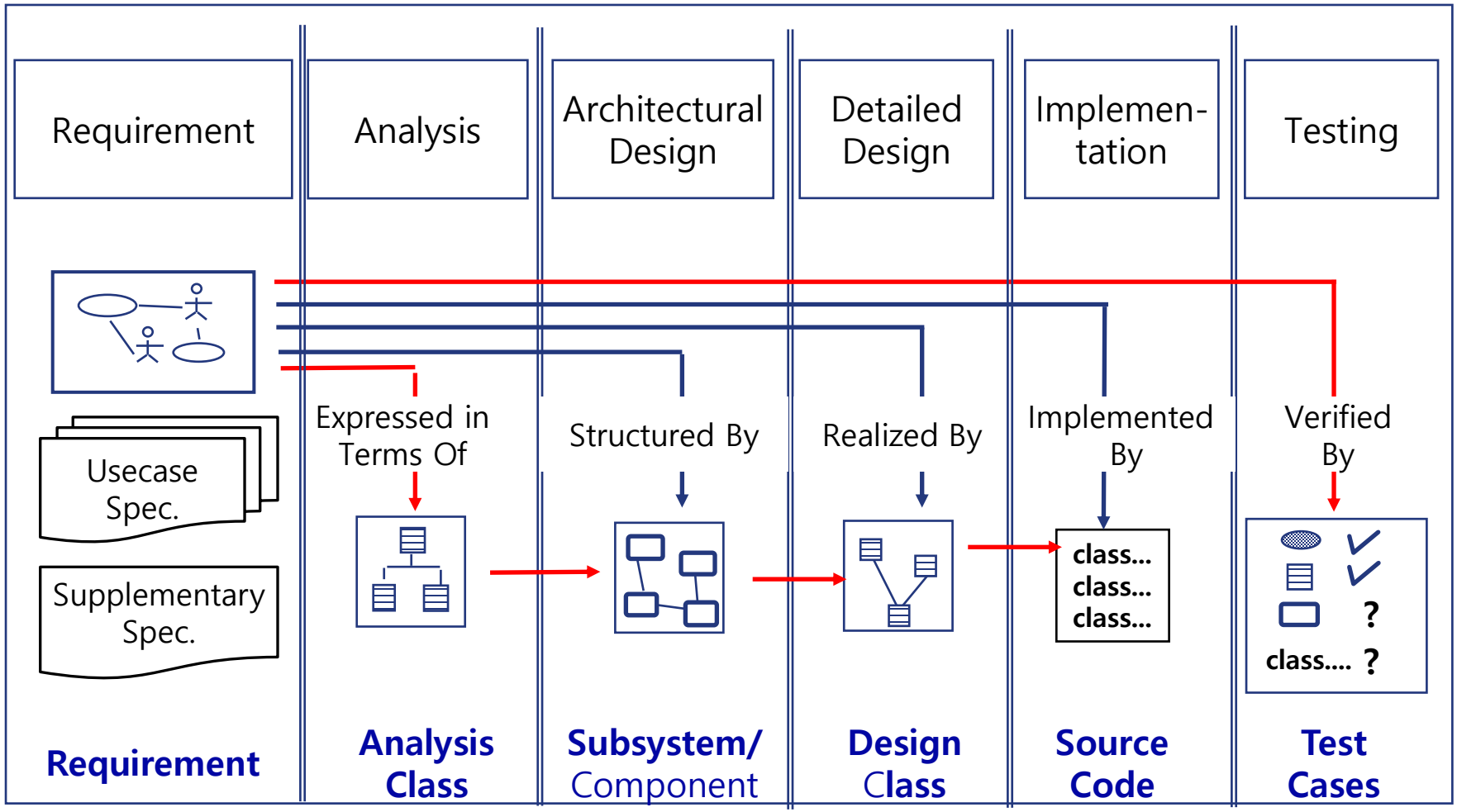


(Sequence diagram)

- **Define Design Class Diagrams**

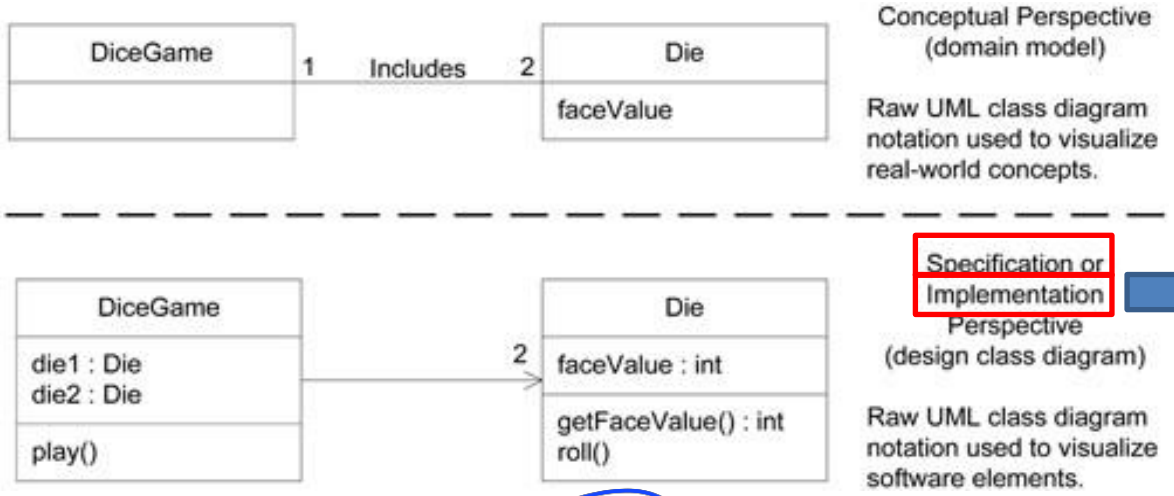


Development Process



Three Perspectives to Apply UML

- The same UML notation may be used for three perspectives and types of models.
  1. Conceptual perspective
     - describe things in a situation of the real world or domain of interest.
  2. Specification (software) perspective
     - describe software abstractions or components with specifications and interfaces, but no commitment to a particular implementation (e.g., not specifically a class in Java).
  3. Implementation (software) perspective
     - describe software implementations in a particular technology (such as Java).



In practice, most software-oriented UML diagramming assumes an implementation perspective and the specification perspective is seldom used for design.